Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/17/2006
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Project Title:	Model-based Support for Crew Activity Planning		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	11/01/2002	**End Date:**	11/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Matessa, Michael (NASA Ames Research Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose to leverage previous research to provide a tool that will allow mission planners to easily construct a CPM-GOMS model of crew activities. The resulting model will enable predictions of the time required to complete common, routine tasks using parameters determined from the existing literature on human performance. The reduction in time to model will now make it feasible to model and simulate performance on several candidate task designs with varying interfaces.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The reduction in time to model will now make it feasible to model and simulate performance on several candidate task designs with varying interfaces.
Task Progress:	Experiment 1- Multifunction Electronic Display System (MEDS) : An experiment was run by the Intelligent Spacecraft Interface Systems (ISIS) lab at NASA Ames Research Center using the Space Shuttle Cockpit Simulator to examine the performance of novice and expert shuttle operators in handling single and multiple malfunctions. The experiment was used to test the hypothesis that reusable templates could predict performance of different participant populations in different workload conditions (low workload for single malfunction trials and high workload for multiple malfunction trials). Five former airlines captains with an average of 15,000 flight hours experience participated in the novice condition. Five current astronauts with a minimum of two years of training participated in the expert condition. Each trial simulated the ascent phase of shuttle operations, starting from launch to Main Engine Cut-Off (MECO) and lasting 8 minutes and 30 seconds of simulated mission elapsed time. In the single malfunction condition, a main engine malfunction was inserted. In the multiple malfunction condition, a combination of helium system regulator failure, General Purpose Computer failure, and coolant system failure malfunctions were inserted. The Flight Data File procedure checklist, which lists all the steps required to recover from the malfunctions, was provided to the subjects during the simulation. Eye fixations were analyzed on regions of interest relevant to solving malfunctions. Conclusions: The experiment tested the hypothesis that reusable templates could successfully predict performance of different participant populations in different experimental conditions. The phrase-reading template made generally good predictions of gaze durations for pilots and astronauts in the single malfunction condition, but there are some noticeable differences. The template represents the process of identifying visual information, but does not include the time to search for the information. The increased gaze time on the switch panel for the pilots relative to the model may represent some search, as the switch panel contains many similar-looking switches. The template also assumes that all information at a location is processed only once. The decreased gaze time on procedures for the astronauts relative to the template may represent the astronaut reading only a relevant subset of the procedure. Likewise, the increased gaze time on procedures for the pilots relative to the template may be due to the pilots reading parts of the procedure more than once. The phrase-reading template made very good predictions of astronaut gaze duration in the multiple malfunction condition, showing that the template generalizes across workload conditions. For the screen touch template, there were not enough data points to evaluate generality across population or workload conditions. When creating the screen touch template, it was assumed that eye fixations were made before touching the screen. Close examination of the data suggests this may be incorrect and that some screen touches may be made without fixations. Experiment 2 – new Cockpit Avionics Upgrade (CAU) : The phrase-reading and screen-touching templates were used to make predictions for an experiment comparing original CAU displays to new CAU in sessions with single or multiple malfunctions. The time for initial finding a malfunction page in the flight data file (FDF) procedures was calculated as gazes at 3.5 (out of 7 possible) tabs plus a gaze to read the malfunction heading for a total of (3.5+1) x 470 = 2115 msec. The time for subsequently finding the correct place on the FDF page after looking at the screen was calculated as an orienting gaze to the FDF and a gaze to a finger keeping place on the page for a total of 2 x 470 = 940 msec. The time for reading information on the screen after looking at the FDF include an extra orienting gaze to the screen. Conclusions: The model assumes the pilots are well-practiced in the procedures. The increased times of the pilots for the first switch throw of procedures may indicate task orientation that is not represented in the model. From analysis of pilot speech, the last switch throw of the single malfunction procedure is sometimes forgotten in the original CAU condition until a later time, and this forgetting is not part of the model. These initial and final increased times are not seen in performance with single malfunction procedures with newCAU, which may assist in task orienting and remembering. In multiple-malfunction procedures, newCAU appears to lessen the increase of initial time due to task orientation. General Discussion: The phrase-reading and screen-touching templates appear to do a good job of predicting behavior. A post-hoc analysis showed that two screen-touching template strategies, one with reading and one without, accounted for the data better than a single strategy. Also, a more detailed screen touch template could include the time required to move the hand from one position to another. This may account for more variance in some of the longer screen touch times. The analyses include only regions of interest that are relevant to solving malfunctions, but there is also a parallel task of monitoring the entire state of the shuttle. Future research will be directed towards modeling the multitasking behavior of the shuttle operators as they solve malfunctions while at the same time monitor the overall state of the shuttle. Publication in earlier years: John, B., Vera, A., Matessa, M., Freed, M, & Remington, R. Automating CPM-GOMS. In: Conference on Human Factors in Computing Systems, CHI 2002 : changing the world, changing ourselves : Minneapolis, Minnesota, USA, April 20-25 / editors, Loren Torveen [et al.] New York : Association for Computing Machinery ; Old Tappan, N.J. : Distributed to nonmembers by Pearson Education, c2002. p. 147-54.
Bibliography Type:	Description: (Last Updated: 05/17/2006)
Articles in Peer-reviewed Journals	Vera AH, John BE, Remington R, Matessa M, Freed MA. "Automating human-performance modeling at the millisecond level." Human Computer Interaction 2005;20(3):225-65. , Apr-2005